

Helping Eyes: Presentation Script

CS3235 Project Review · Title, Literature Survey and Problem Statement

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SLIDE 1

Title

Speaker: Neeraj P Rao (1RVU23CSE303)

Good [morning / afternoon]. I'm Neeraj, and with me are Parveen and Misha. Our project is called Helping Eyes, a reading assistant for people who are blind or have low vision.

The idea in one line: you hold a medicine strip, a food label, a letter or a book in front of an ordinary webcam, and instead of the system reading every single word aloud, you ask for exactly what you need. For example, "When does it expire?" or "Read only the dosage". You then hear a short spoken answer.

Two design choices sit behind the title. First, the text is recognised on the device itself, using Apple's Vision framework. Second, the questions are answered by a language model that runs locally, Qwen 2.5 7B through Ollama, so the captured text never has to leave the laptop.

In this review we will cover three things. First, the literature we surveyed: nineteen papers, plus the WHO World Report on Vision. Second, the research gaps those papers leave open. Third, our problem statement and the scope of the project.

Parveen will begin the literature survey with the classic camera-and-speech reading aids.

SLIDE 2

Literature survey 1 of 4: Camera, OCR and speech reading aids

Speaker: Parveen (1RVU23CSE330)

Thank you, Neeraj. I'll start with the most common family of work: systems that take a picture of printed text, run OCR on it, and speak the result.

Sharma and colleagues, in 2014, built a classic page reader. It finds text regions with connected-component analysis, recognises the characters with OCR, and speaks them with concatenative speech synthesis. That is the basic pipeline most later systems follow.

Ezaki, Bulacu and Schomaker, back in 2004, focused on the first step: finding text in natural scenes, explicitly as part of a system for visually impaired people. Their contribution is detection. They do not go on to answer anything about that text.

Yi, Tian and Ardit, in IEEE/ASME Transactions on Mechatronics, tackled product labels on hand-held objects. The user shakes the object, and the motion tells the system where the region of interest is. It is a clever idea, but it needs that deliberate gesture, and it then reads out all the text it finds.

The two recent papers, Gawande and colleagues in 2023 and the Smart Reader paper at APSIT 2025, show the same idea still being built today as dedicated devices that convert print to speech.

The common thread is this: all five treat reading as a read-out. For a medicine pack with hundreds of words, the user has to listen to everything to find the one line they need. Misha will now cover wearables, commercial apps, and the problem of aiming the camera.

SLIDE 3

Literature survey 2 of 4: Wearables, apps and aiming without sight

Speaker: Misha (1RVU23CSE269)

Thanks, Parveen. The second group of papers looks at the device and at a problem that is easy to overlook: pointing a camera at text when you cannot see it.

FingerReader, from Shilkrot and colleagues at CHI 2015, is a camera worn on the finger. It reads the line directly under the finger and gives audio and haptic cues to keep the user on the line. It works, but it needs custom hardware, and the user still has to scan the text line by line.

Granquist and colleagues compared two commercial aids, the OrCam MyEye wearable and Microsoft's Seeing AI app. Both were above 95 percent accurate on flat, plain documents. On formatted text on curved surfaces, though, accuracy dropped to between 13 and 57 percent. That tells us real-world packaging is much harder than a printed page.

The next three papers are about aiming. Jayant and colleagues surveyed how blind people actually use cameras and why their photos often come out blurred or cut off. Cutter and Manduchi studied how blind users hold a phone over a document and tested spoken guidance to

help them get an OCR-readable picture. Neat and colleagues compared three mobile OCR approaches, including one that vibrates when text is detected and waits for the phone to be held steady before taking the picture.

Our takeaway: guidance helps a lot, but in these systems it ends once a good photo is taken. After that, the app still reads out all the text it found. Neeraj will now cover the recognition engines and models our system builds on.

SLIDE 4

Literature survey 3 of 4: Recognition, layout and language models

Speaker: Neeraj P Rao (1RVU23CSE303)

Thanks, Misha. This slide covers the building blocks: how text is recognised, how a page is put in reading order, and the language model we answer questions with.

Nagy and Seth introduced the recursive XY-cut in 1984. You split a page at its widest empty gap, either between columns or between blocks, and repeat on each part until nothing can be split. The result is the blocks in reading order. It is still a standard layout method, but it was designed for flat scanned pages, not camera frames, and not for an open book showing two pages at once. We adapt it for our book reading mode.

Smith's 2007 paper describes Tesseract, the best-known open-source OCR engine. It works very well on clean scans, but images from a hand-held camera need extra processing to get good results.

PP-OCR, from Du and colleagues, is a lightweight OCR system whose English and Chinese models are only a few megabytes. That makes it practical on an ordinary CPU. The web version of our project uses RapidOCR, which runs PaddleOCR models like these. On the Mac, we use Apple's Vision framework instead.

Finally, the Qwen 2.5 technical report describes the open-weight model family we use: the 7B instruct model on the desktop and the 3B model in the web version. The limitation matters: a general model will happily answer from its own knowledge. For a medicine label, that is exactly what we must prevent, so every answer has to be grounded in the captured text.

Parveen will now cover work on answering questions about what the camera sees.

SLIDE 5

Literature survey 4 of 4: Answering questions about what the camera sees

Speaker: Parveen (1RVU23CSE330)

Thanks, Neeraj. The last group moves from reading text to answering questions about it, which is the core of our project.

VizWiz, by Bigham and colleagues at UIST 2010, let blind users take a photo, record a spoken question, and get answers from crowd workers on the web within about a minute. It proved that blind people want to ask questions, not just hear descriptions. But it depends on paid human workers, and every photo leaves the user's phone.

Gurari and colleagues turned those questions into the VizWiz dataset at CVPR 2018: over 31,000 visual questions from blind users, each with ten answers. They found that many photos are poor quality and that many questions simply cannot be answered from the image. That supports the need for good capture before answering.

Gonzalez Penuela and colleagues, at CHI 2024, ran a two-week diary study with 20 blind and low-vision people using an app built on a multimodal large language model. Users found it useful, but rated satisfaction only 2.76 out of 5 and trust 2.43 out of 4.

Two papers explain why trust is low and what helps. Lewis and colleagues introduced retrieval-augmented generation: base the model's answer on retrieved text instead of its memory. Ji and colleagues surveyed hallucination, where models generate fluent text that the source does not support. For a dosage or an expiry date, one invented detail can cause real harm.

Misha will now bring these four themes together into the research gaps.

SLIDE 6

Research gaps and relevance

Speaker: Misha (1RVU23CSE269)

Thanks, Parveen. Putting the nineteen papers together, we see five gaps, and each one maps directly to a part of our project.

One. Almost every reading aid treats reading as a read-out. Papers 1, 3, 4, 5 and 10 all end with the system speaking whatever text it found. Our response is question-driven reading: the user can still say "read everything", but can also say "read only the dosage" or ask "when does it expire?" and hear a short answer.

Two. Aiming a camera without sight is hard, as papers 8, 9 and 10 show. We run fast text detection about ten times a second, give spoken hints like "Move closer" or "Move left", and capture automatically once the text is held steady.

Three. Many solutions need dedicated hardware, like FingerReader or the device in paper 4, or send the photo to other people, as VizWiz does. Our system runs on an ordinary Mac with a webcam, and both OCR and the language model run on the device.

Four. Language models can invent details, and users do not fully trust them, as papers 17 and 19 show. We instruct the model to answer only from the captured text and to say when the answer is not there. Expiry dates are compared with today's date in code, not by the model. A web lookup is offered only when the text does not have the answer, and it runs only if the user agrees.

Five. Classic layout methods assume flat, single-page scans, and accuracy falls on formatted or curved print. Our book mode detects page turns, splits a two-page spread at the spine, reads the printed page numbers and running titles, and orders paragraphs with an adapted XY-cut.

Neeraj will now state the problem precisely.

SLIDE 7

Problem statement

Speaker: Neeraj P Rao (1RVU23CSE303)

Thanks, Misha. Here is our problem statement.

Visually impaired people cannot independently get specific information from printed text: medicine labels, food packaging, letters, bills and books. The tools available today are either expensive dedicated devices, or phone apps that read every recognised word aloud and give little help in pointing the camera at the text.

So the problem we are addressing is to build an affordable assistant that runs on an ordinary laptop and webcam, guides the user to capture the text, and then speaks only the information the user asks for, grounded in the captured text.

The three boxes below turn that statement into requirements we can test.

Guide the capture: the system detects text live and gives spoken hints until the text is fully in view, then captures automatically after a short hold, with no button press needed.

Answer what is asked: the user can ask for a full read-out, for one part such as the ingredients or the directions, or ask a question, by voice or by typing.

Never invent details: answers come only from the captured text, and when the text does not say, the system says so. Exact facts like expiry dates are computed in code, because we observed the 7B model getting date comparisons wrong.

Parveen will explain why this problem matters.

SLIDE 8

Problem statement: Significance

Speaker: Parveen (1RVU23CSE330)

Thanks, Neeraj. Why does this matter?

According to the World Health Organization's World Report on Vision, published in 2019, at least 2.2 billion people worldwide have a near or distance vision impairment. The WHO also notes that the burden falls more heavily on low- and middle-income countries, where expensive dedicated reading devices are least affordable.

Beyond the scale, we see three reasons this problem is significant.

Safety. Much of the printed text people need is safety-critical: medicine dosages, warnings, allergens and expiry dates. Reading these wrongly, or not being able to read them at all, can cause real harm. That is why our system is strict about answering only from the text, and why expiry dates are checked in code.

Independence. Today, many visually impaired people ask a family member, or use services like VizWiz that send the photo to a sighted person. A system that answers on its own lets people read their own post and their own medicine without depending on someone else.

Affordability. Our desktop system needs no special hardware: a Mac with a built-in or USB webcam is enough. For other devices, there is a browser version.

Misha will now define exactly what is in and out of our scope.

SLIDE 9

Problem statement: Scope

Speaker: Misha (1RVU23CSE269)

Thanks, Parveen. To finish the problem statement, here is exactly what the project covers.

In scope, on the left. Live text detection from the webcam, with spoken guidance and automatic capture. Requests by voice or by typing: read everything, read only one part, or ask a question. Answers come from a local Qwen 2.5 7B model and are spoken sentence by sentence as they are generated, so the first sentence arrives quickly. Expiry dates are checked in code, and a web search is offered only when the text lacks the answer and runs only with the user's consent. Book mode reads a book page by page: it detects page turns, splits two-page spreads, announces page numbers and orders paragraphs. Finally, a browser version runs the same answering logic on a server, for people without a Mac.

Out of scope, on the right. This is a reading assistant, not a navigation aid, so obstacle detection and general object recognition are not part of it. We do not read Braille, and handwriting is not a design target. The system never gives medical advice beyond what is printed on the item; web answers are clearly marked "According to the web" and tell the user to check with a pharmacist or doctor. Spoken requests currently use Google's online speech recognition, so offline speech input is out of scope, though typed requests work offline. And the browser version has no live guidance or automatic page-turn detection.

Neeraj will now take you through the references.

SLIDE 10

References [1]-[10]

Speaker: Neeraj P Rao (1RVU23CSE303)

These are the first ten references, the ones from the first two literature slides.

References 1 to 5 are the camera, OCR and speech reading aids Parveen covered, from Ezaki's 2004 text-detection work up to the Smart Reader paper at APSIT 2025. References 6 to 10 are the wearables, commercial apps and camera-aiming studies Misha covered.

They are published in IEEE, ACM and journal venues, including CHI, ASSETS, DocEng, IUI, ICPR and IEEE/ASME Transactions on Mechatronics, and they span 2004 to 2025. That lets us show both the long history of the read-out approach and the fact that it is still being built today.

Parveen will cover the remaining references and close.

SLIDE 11

References [11]-[20] and closing

Speaker: Parveen (1RVU23CSE330)

These are the remaining references. 11 to 14 are the building blocks Neeraj covered: the XY-cut layout algorithm, the Tesseract and PP-OCR recognition systems, and the Qwen 2.5 model family. 15 to 19 are the visual question answering and language model papers I covered: VizWiz, the VizWiz dataset, the 2024 diary study of an LLM app, retrieval-augmented generation, and the hallucination survey. Reference 20 is the WHO World Report on Vision, the source of the 2.2 billion figure.

To summarise the review: nineteen papers show that reading aids have mostly treated reading as a read-out, that aiming the camera without sight is a real problem, and that language-model answers must be grounded to be trusted. Helping Eyes addresses these gaps with guided capture, question-driven answers from the captured text, and a book mode for multi-page reading.

Thank you. We are happy to take questions.